

Element I: Testing, Data Collection, and Analysis

Table of Results:

Test Name	Structure	Detection and Notification Time	Time for Suppression	Priority of test
Test Results	Room 1 12x12 Room 2 6x12	Gas threshold for MQ2 sensor - 300 ppm IR sensor - activated(1)	Fire must be suppressed in less than 5 minutes	Structure- to have all the dimensions right on target Detection and Notification Time- The detection and Notification system activates in significantly less than 30 seconds Time for Suppression- The suppression system activates simultaneously with the notification system and suppresses the fire in under 2 minutes
Number of Trials	8/12 trails	7/7 trials, and we had to see if we could detect the fire under 30 seconds	In 6/7 trials, we had to see if the fire could be suppressed in under 5 minutes consistently	

Observations:	I found out that the structure of a building plays an important role. A building's structure significantly impacts how fires behave and spread, influencing both the fuel load available for combustion and the structural integrity of the building.	The first two trials failed, but after adjusting the gas sensor activation thresholds, the next three trials succeeded, albeit with inconsistent results. Further code adjustments and improvements to the Arduino connection allowed for activation in under 30 seconds, depending on fire intensity.	In the first trial, we could only get the pump to work manually. After adding a relay module and connecting it to the Arduino and a 9V battery and we were able to get it to under 2 minutes in a couple more trials and modifications	
Pass/ Fail	Pass	Pass	Pass	

Summary of the testing data:

Verbal Summary: Our table of results displays and explains what we did for our Structure, Detection and Notification Time, and Time of Suppression, and their trials, tests, and observations we had. This describes what we did and how we designed and constructed our Fire Day project.

Tables and Graphs:

Trial number	Detection and Notification Time(secs)
1	N/A
2	N/A

3	62
4	53
5	67
6	28
7	26

Detection and Notification Time(secs) vs. Trial number

